



Project information – OP4512		
Project Number	CC-221695 (OPT906637)	
Customer Name	IIT Guwahati	
Software Version Information		
MATLAB Version	R2021B	
RT-LAB Version	2024.1.3.44	
HYPERSIM Version	N/A	
Other software (if applicable)	N/A	
Additional information		
External Equipment S/N	N/A	
Date of Execution (DD/MM/YYYY)	18/04/2024	
Executed by	M.H	
Target Information		
OS	OPAL-RTLinux 3.5.4	
Number of CPU(s)	1	
Number of Core(s)	4	
Target Communication	UDP/IP	
SFP Connection Type	N/A	
Test Steps		
Test Step	Expected result	Status
Verify that the Unit Under Test matches the requested configuration (OS configuration, number of CPUs, IO presence, bitstream, etc)	Approved Visio should match the list of items found in the P.O.	Pass ☒ N/A ☐
Verify that the system powers on properly, that it pings	- NC1 LED on front starts to blink (when connected on network) - Following LEDs are On - CFG OK, PWR ON, MASTER/SLAVE (green when in master mode, yellow in standby)	Pass ☒ N/A ☐
PCIe detection of the FPGA	- Board Identification Module (bim) detects the proper FPGA (KINTEX 7) and correct board ID - PCIe Gen2 / X4 detected	Pass ☒ N/A ☐
Verify the functionality of the IO signals on the DB37 connectors	All modules and channels are functional	Pass ☒ N/A ☐
Verify the functionality of the IO signals on the DB9 connectors	All modules and channels are functional	Pass ☒ N/A ☐
Validate that the synchronization pulse is present when model is running.	TX LED is on and pulse is measured.	Pass ☒ N/A ☐
Validate control of USER 1 and USER 2 LEDs	LEDs are controllable via RT-LAB model	Pass ☒ N/A ☐
Increase/decrease the time factor until the model can run without overruns (<i>stretched step in HYPERSIM</i>) and Inputs/Outputs remain functional. Test performed to validate CPU performances.	The minimum step size is $T_s = 7\mu s$	N/A ☐
Model information		
RT-LAB test model name (.slx)	IO_test	
Bitstream configuration file name (.bin)	TE0741_4-EX-0001-3_5_0_817-eHSx128m_Gen4_DBD8D8C022QEIO-4F-23	
Additional Comments		
System delivered has been tested using OPAL-RT TECHNOLOGIES I/O modules with serial numbers listed in tables that follow.		
Test Result Approval		
The undersigned acknowledge they have reviewed the Test Report and agree with the approach it presents. Changes to this Test Report will be coordinated with and approved by the undersigned or their designated representatives.		
Signature:	Mina Hazian	Date (DD/MM/YY)
		18 / 04 / 2024

**OP5330-1 Test Description**

- Verify the functionality of all channels (CH00..15).
- Tested in loopback configuration with OP5340 (Analog input module)
- Verify the output and error values at multiple output ranges (-16, 0 and 16 V) with maximum acceptable offset (at 0V input) and error (at $\pm 16V$ error) values to be at 35 mV.

Test Information

Test Performed on	OP5330-1 rev 3.1
Product Number	126-0104
Quantity delivered with system	1
Serial Number	23-7158
Tested on system by	M.H
Comments	

OP5342 - Test Description

- Verify the functionality of all channels (CH00..15).
- Tested with external source providing voltages of -18, 0 and 18 V
- Verify the input and error values at multiple output ranges (-18, 0 and 18 V) with maximum acceptable offset (at 0V input) and error (at $\pm 18V$ error) values to be at 15 mV.

Module Information

Test Performed on	OP5342 rev 2.0
Product Number	126-0559
Quantity delivered with system	1
Serial Number	22-6855
Tested on system by	M.H
Comments	

OP5369- Test Description

- Verify the functionality of all channels (CH00..31).
- Validate that the card is functional in Din mode and Dout mode.
- Validate VTH and VTL
- Validate the functionality of all voltage level indicator LEDs on the board.
- Validate Dout 5V when No User present

Module Information

Test Performed on	OP5369 rev 4.0
Product Number	126-0640
Quantity delivered with system	2
Serial Number	24-1967, 24-1968
Tested on system by	M.H
Comments	

OP5969-1 - Test Description

- Verify the functionality of all channels (CH00..05).
- Validate the differential RS-422 digital output channels in loopback configuration with Pulse-Width Modulator generated from Digital Out module under test and read from the differential RS-422 digital input channels.
- Validate the functionality of all voltage level indicator LEDs on the board.

Module Information

Test Performed on	OP5969-1 rev. 1.1
Product Number	126-0536
Quantity delivered with system	1
Serial Number	24-0278
Tested on system by	M.H
Comments	



STARTER KIT INSTALLATION

KIT CONTENTS



The starter kit box MAY NOT include all the following hardware, depending on your system.

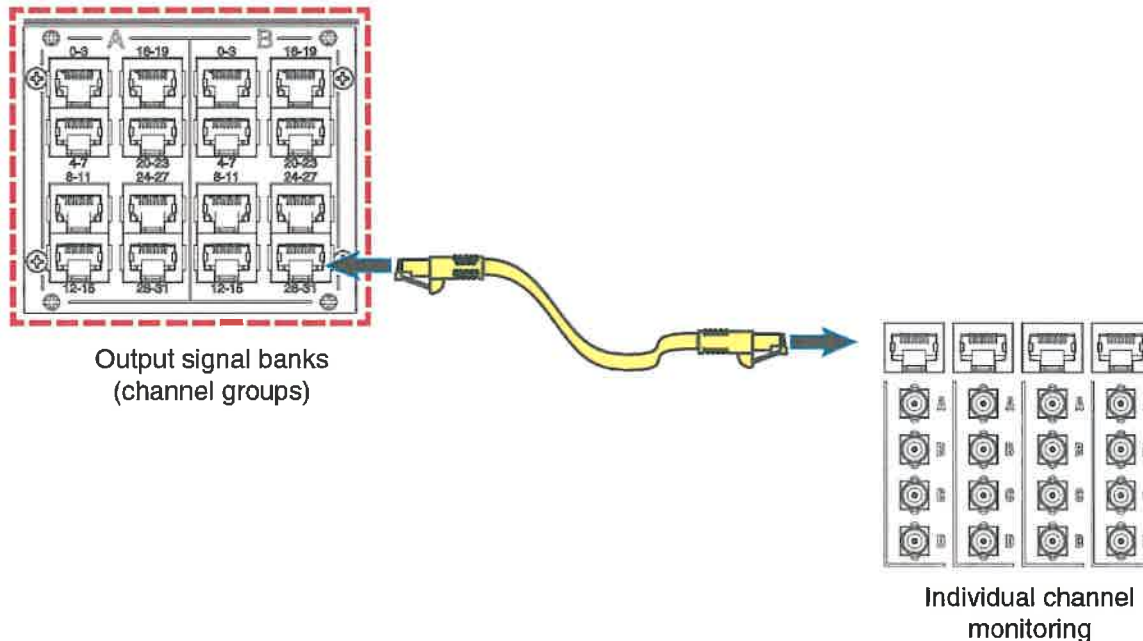
Not all parts listed here are included with OPAL-RT's OP4500 series of products.

Consult the product user manual for more detailed information.

CONNECT RJ45 CABLES

Four RJ45 cables (red, green, black and yellow) are used to provide a connection from output connectors from a block of channels to a series of connectors on an OPAL-RT simulator to monitor each channel in the block. Note that not all OPAL-RT simulators contain these features. Always refer to your specific simulator user manual. (The blue cable is used to establish a network connection.)

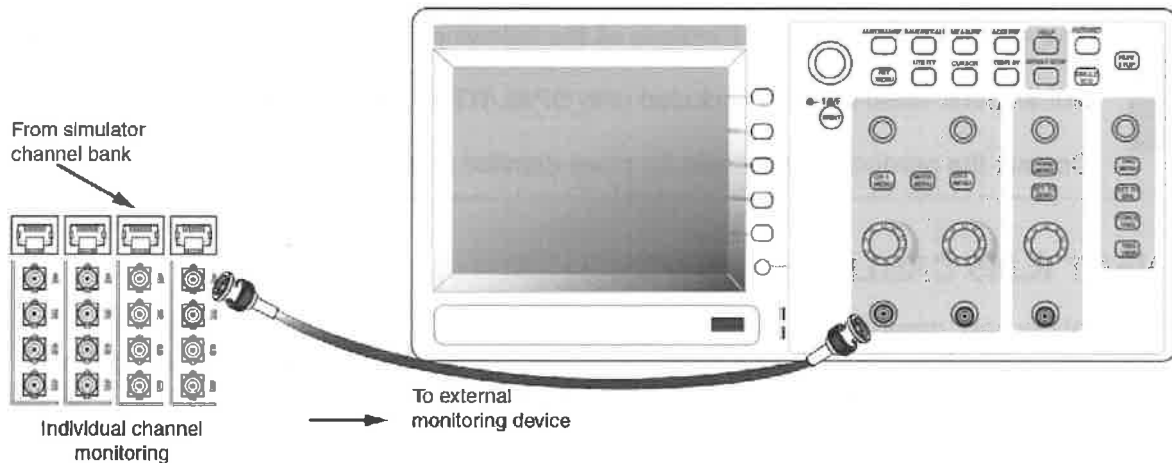
OPAL-RT SIMULATOR CONNECTORS



Refer to your simulator user manual for detailed use instructions.

CONNECT MINI BNC TO BNC

These cables establish connections between OPAL-RT hardware and external monitoring devices.

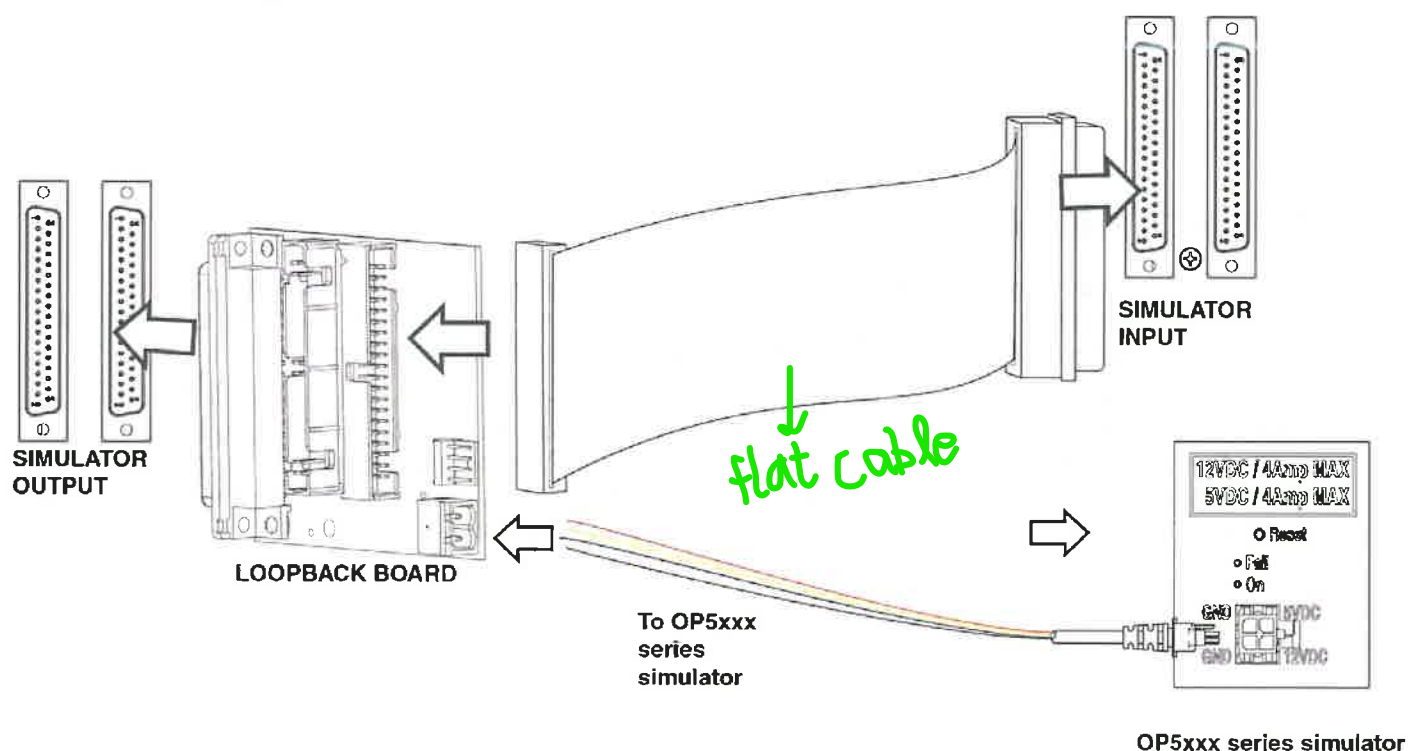


Connect a mini-BNC cable to each jack (C) and connect the other end of the cable to the desired monitoring device.

CONNECT THE LOOPBACK KIT

The loopback kit included in the starter kit allows users to test system signals. Connection is made easy with a custom flat cable that matches OPAL-RT standard DB37 pin assignments.

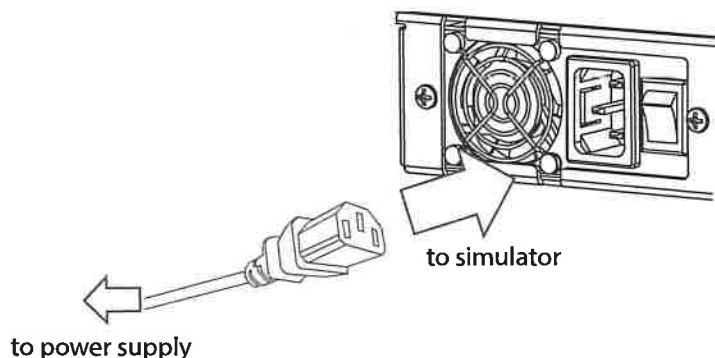
1. Connect one end of the flat cable to the loopback board
2. Connect the loopback board to the simulator DB37 Output signal
3. Connect the other end (DB37 connector) of the flat cable to the simulator DB37 Input signal
4. Connect the VUser (required to preserve isolation) from the loopback board to an OPAL-RT simulator (5000 series only).



Although some boards do provide a power source, using that would compromise the isolation. OPAL-RT recommends using external power source: the user must connect the power wires (provided) to either a 5 or 12 V power source. Make sure that the switch on the loopback board is set for the proper voltage.

CONNECT THE POWER CABLE

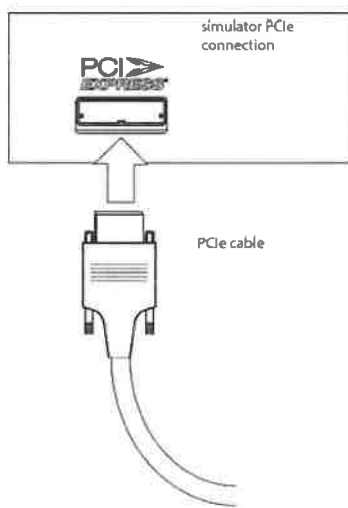
Connect one end of the power cord to the simulator and the other end to the power source (wall plug, extension, etc.).



CONNECT THE PCIE CABLE

If your device came as part of a full system delivery, simply connect the PCIe cable between the simulator and the device PCIe connector in the back of the unit. Make sure to connect the cable end labelled 'host' to the simulator, and the cable end labelled 'target' to the device.

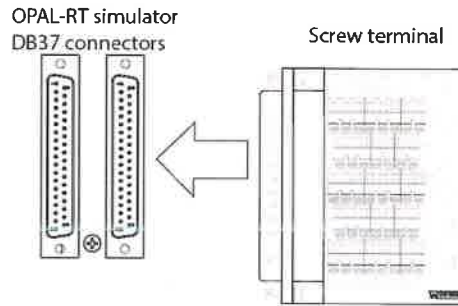
If the chassis was purchased as a separate unit, it is delivered with a PCIe cable and a PCIe interface board. This board must be installed in your OPAL-RT simulator. Contact OPAL-RT Support for instructions to install the PCIe adaptor board in the simulator. Then connect the PCIe cable from the PCIe connector at the back of the chassis to the simulator. Make sure to connect the cable end labelled 'host' to the simulator, and the cable end labelled 'target' to the chassis.





CONNECT THE BREAKOUT TERMINAL

The breakout board provided allows users to access each pin of the DB37 separately through the screw terminals. Simply insert the breakout board DB37 connector onto the desired DB37 connector on the simulator.



Refer to the simulator or board user manual for exact pin assignments

CONNECT THE SYNCHRONIZATION CABLE

FIBER OPTIC CABLE

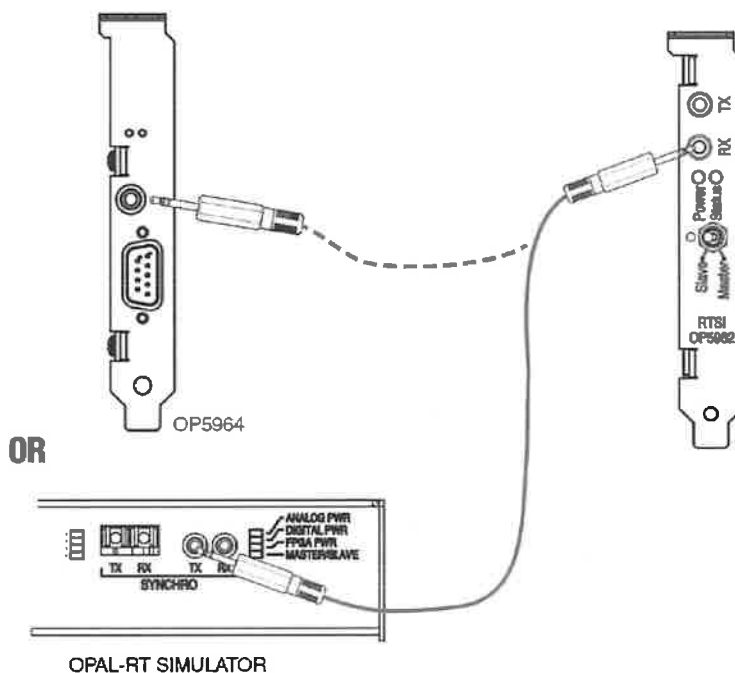
The fiber optic cable is used to establish synchronization connections between simulators (as shown below) or to provide loopback for boards equipped with fiber optic connectors.





AUDIO CABLE

Note that some systems do not have fiber optic connections and use standard audio cable and jacks (the connectors to the right of the fiber optic connector in the image) to make synchronization connections.

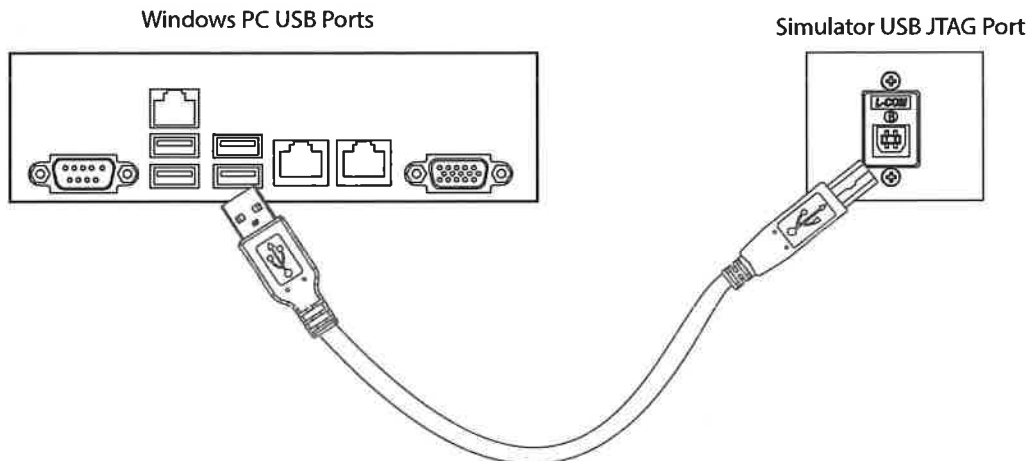


Refer to the system description document (in the Integration Binder) for more detailed connection instructions.

CONNECT THE USB A TO USB B CABLE

The USB(A)-USB(B) cable provides the connection required to flash the FPGA using the JTAG port.

Connect one end to a Windows PC USB port and the other end to the USB JTAG port on the OPAL-RT simulator, then follow the instructions to flash the FPGA.



CONTACT

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Simulator EMC cable recommendations

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History			
	Author		Changes Log Book
Version	Name	Date (MM/DD/YYYY)	Modification subject
1.0	Wilfrid Gelin François Lemieux	03/30/2022	Initial release

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1. CONTEXT

This document presents recommendations of good cabling design to ensure electro-magnetic compliance (EMC) of the simulator. Failing the recommendations might result of energy transfer in electro-magnetic interference (EMI) that can disturb the equipment in the neighbor environment.

2. CABLE COMPONENT ITEMS

1. Ferrite clamp of proper circumference
2. Twisted pairs for signal and power
3. Shield

3. ANALOG CABLE ASSEMBLY

The D-sub37 cable used for the analog IO must follow the recommendations:

1. Data should be cabled in twisted pairs.
2. Cables should have a shield around the data.
3. One shield per type of IO and direction.
4. The shield should be connected to the connectors' shell/casing at both ends.
5. The shells/casings should be connected to the chassis earth ground on both equipment.
6. The total cable length should be less than 3 m (9 feet).

4. DIGITAL CABLE ASSEMBLY

The D-sub37 cable used for the digital IO must follow the recommendations:

1. Data should be cabled in twisted pairs.
2. Cables should have a shield around the data.
3. One shield per type of IO and direction.
4. The shield should be connected to the connectors' shell/casing at both ends.

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5. The shells/casings should be connected to the chassis earth ground on both equipment.
6. The total cable length should be less than 3 m (9 feet).
7. For Digital Output cables, the VUser_RTN signal should be connected to the shell/casing of the cable with a 100nF capacitor on both ends.
8. The cable should have a ferrite clamp on the Digital Output side.
 - a. During EMC certification, Opal-RT used 3 clamps (Laird-Signal Integrity Products #28A2913-0A2) mount like in figure 1.

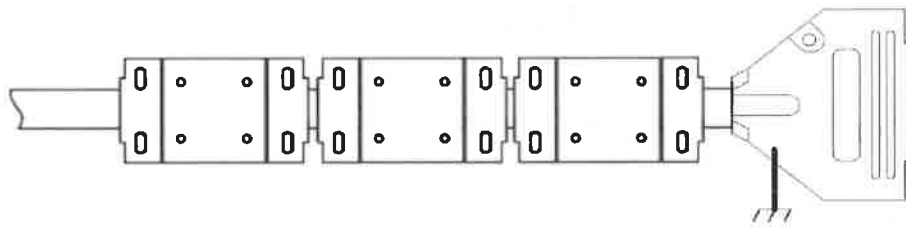


Figure 1: Simulator DB37 digital output connector side

5. POWER CABLE ASSEMBLY

The power cable used for the VUser in DOut cards must follow the recommendations:

1. Power connections should use twisted pairs.
2. Cable should have a shield around the twisted pairs.
3. One shield per type of IO and direction.
4. The shells/casings should be connected to the chassis earth ground on both equipment.
5. The total cable length should be less than 3 m (9 feet).
6. The cable should have a ferrite clamp on the Digital Output side.
 - a. During EMC certification, Opal-RT used the clamp (Laird-Signal Integrity Products #28A2736-0A2) mount with multiple turns like in figure 2.



Figure 2: Power cable for VUser

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